

HackPulse

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PROBLEM STATEMENT

Title: The Hidden Gap in Elderly Care: Sarcopenia Goes Undetected

- Sarcopenia (age-related muscle loss) leads to falls, fractures, long hospital stays and loss of independence
- Confirming it needs DEXA or CT, which most Tier-2 and Tier-3 hospitals do not have
- Patients are diagnosed only after a fragility fracture or repeated admissions
- Two routine tests already exist in every clinic: knee X-ray and handgrip test, but nobody combines them
- Osteoporosis often goes unnoticed in the same patients
- Bottom strip (one line): The data is already there. What is missing is the intelligence to connect it.



Solution : SARCO - SCAN

SarcoScan: Screening Sarcopenia and Osteoporosis from a Routine Knee X-ray

- AI-powered screening tool for web and mobile, running fully on-premise
- Input: one knee AP X-ray + handgrip strength + basic details (age, sex, BMI)
- Output: sarcopenia stage (No / Possible / Probable / Severe) + osteoporosis risk (Low / Moderate / High) + one-page report
- Based on AWGS 2019 criteria, suited to Indian patients
- No DEXA, no CT, no new equipment

Why are we different :

1. Uses data already collected, so zero extra cost per patient
2. Fusion of image + grip beats grip alone
3. Explainable output (overlay + measured values vs cutoffs)



USER WORKFLOW

From Patient Walk-in to Report in Under 5 Minutes

1. Register patient
2. Enter clinical inputs and handgrip readings
3. Upload knee X-ray (or auto-receive from PACS)
4. Image quality check
5. AI analysis (about 5 seconds)
6. Risk results with explainable overlay
7. Doctor review and override
8. PDF report + follow-up trend



Technician

AI System

Doctor

1



Register
Patient

2



Enter Handgrip
and Clinical
Inputs

3



Upload
Knee X-ray

4



Quality
Check

5



AI Analysis
in 5 sec

6



Risk Results
and Overlay

7



Doctor
Review

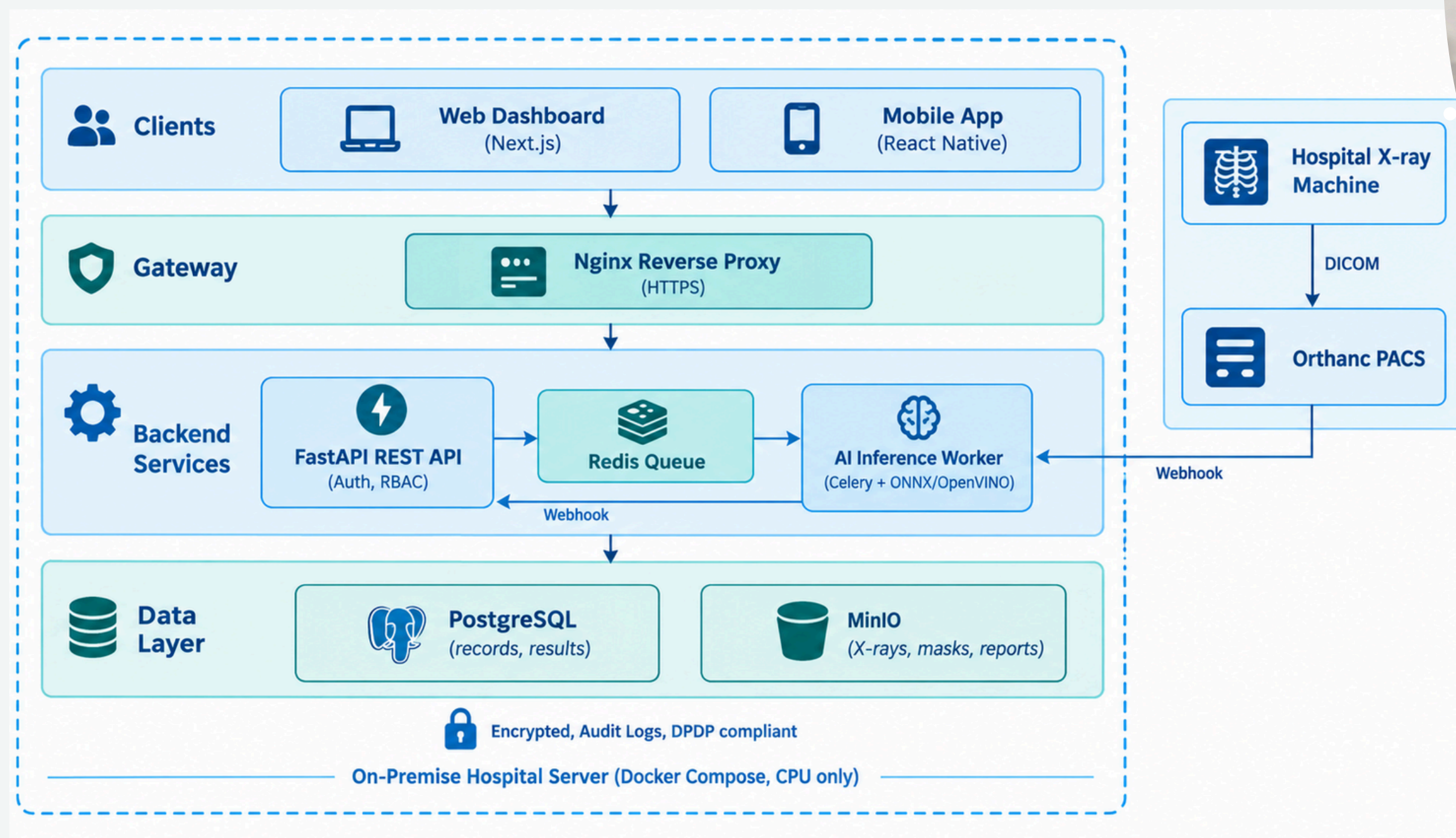
8



PDF Report
and Trend

SYSTEM ARCHITECTURE

System Architecture: Lightweight, On-Premise, Hospital-Ready

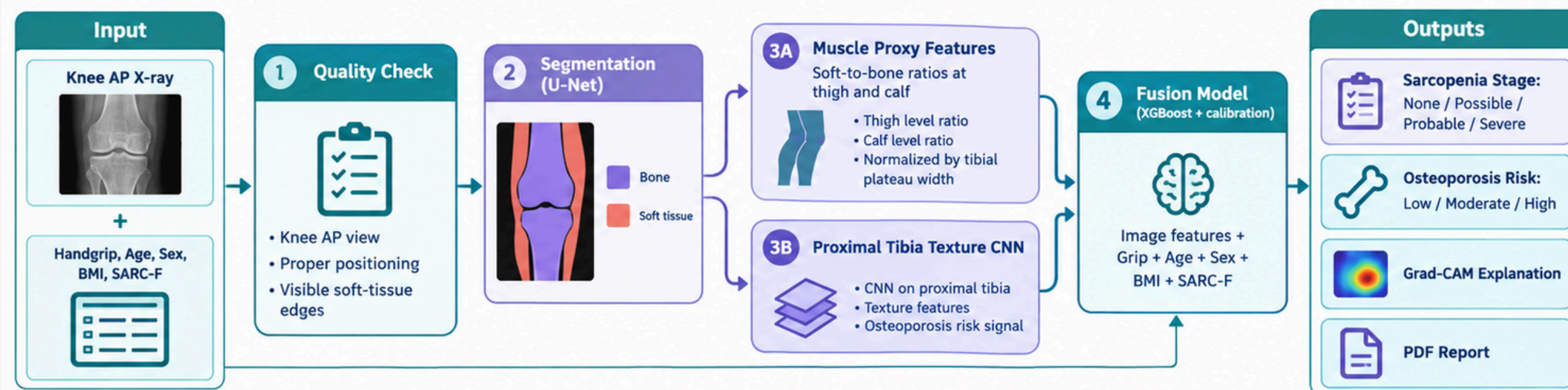


TECHNICAL APPROACH

From Patient Walk-in to Report in Under 5 Minutes

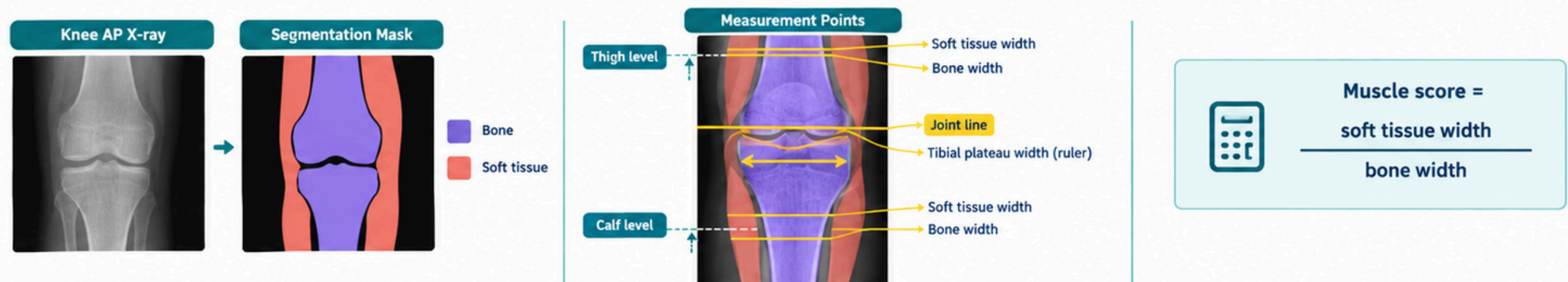


AI Pipeline for Sarcopenia & Osteoporosis Risk



Honest Note: A plain X-ray is a weak muscle signal on its own, so we fuse it with grip strength and BMI. Our claim: **fusion beats grip alone.**

How We Measure Muscle & Bone (Example: Knee AP View)



Key Features and Tech Stack

Features :

- Role-based login (Technician, Doctor, Admin)
- Patient registration and visit history
- X-ray upload with automatic quality check
- Segmentation overlay and muscle measurements
- Sarcopenia stage and osteoporosis risk
- Explainable results (Grad-CAM)
- Standardized PDF report + FHIR export
- Longitudinal trend graphs
- Works offline on local network

 Frontend	   React Next.js Tailwind
 Backend	   FastAPI Celery Redis
 Database	  PostgreSQL MinIO
 AI	      Python PyTorch MONAI XGBoost ONNX OpenVINO
 Deployment	  Docker Compose Nginx
 Integration	  Orthanc (DICOM) FHIR

VALIDATION AND IMPACT

Validation plan:

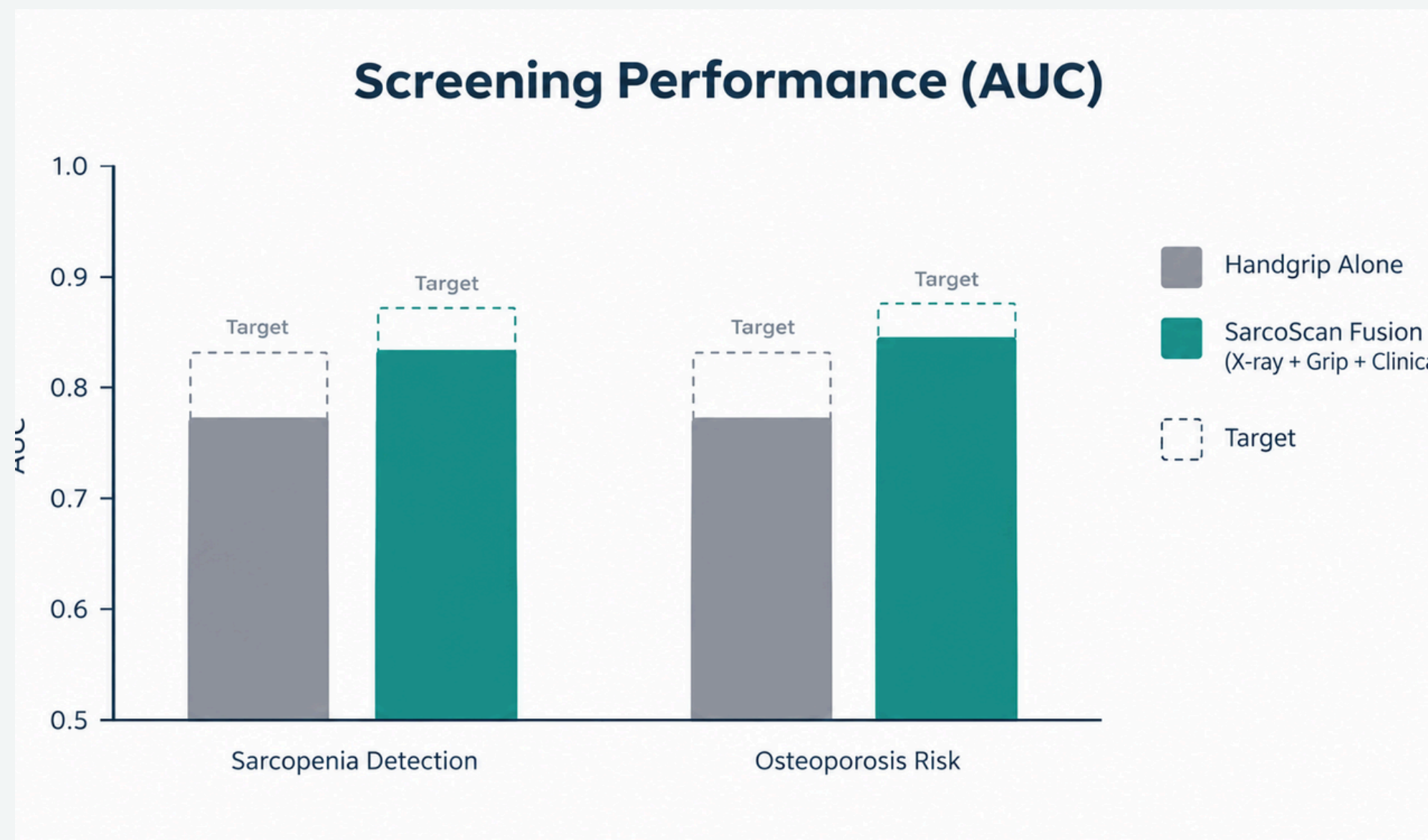
Data: OAI knee radiographs (thigh MRI subset for weak muscle labels) + partner-hospital X-rays with paired DXA

Primary test: AUC of fusion model vs grip alone (DeLong test)

Also: net reclassification improvement, calibration plots, sensitivity at screening threshold

Patient-level splits and external-site validation

Prospective OPD pilot to measure time per patient





FEASIBILITY, ROADMAP AND FUTURE SCOPE

FEASIBILITY :

- Runs on one standard i5 / 16 GB computer, no GPU
- Uses existing X-ray machines and PACS, with no new hardware or consumables
- Data stays on-premise, DPDP Act 2023 aligned
- Fits in the existing OPD workflow

FUTURE SCOPE :

- Improve muscle estimation with DEXA-labelled training data
- Add fall-risk prediction
- Support other X-rays (hip, hand)
- Regulatory path (CDSCO) for clinical use

ROADMAP :

PHASE	PLAN
PHASE 1	MVP: upload, segmentation, fusion score, report, trends
PHASE 2	Orthanc and FHIR integration, KL grade, Hindi UI
PHASE 3	Prospective pilot and external validation
PHASE 4	ABDM / ABHA integration, multi-hospital rollout